

SMALL SAMPLE PERFORMANCE OF CONFIDENCE INTERVALS FOR A TRUNCATION PARAMETER

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Summary: Let $\{X_n, n \geq 1\}$ be a sequence of independent and identically distributed random variables with common cumulative distribution function $F_\theta(x)$, where θ is a finite left truncation parameter. In this paper fixed-width confidence intervals with prescribed coverage probability are investigated.

1. INTRODUCTION

Let $\{X_n, n \geq 1\}$ be a sequence of independent and identically distributed (i.i.d.) random variables with common cumulative distribution function (c.d.f.) $F_\theta(x)$, where θ is a finite left truncation parameter (that is $F_\theta(\theta) = 0$). The problem is to construct a confidence interval for θ having prescribed width $d > 0$ and coverage probability $1 - \alpha$ ($0 < \alpha < 1$). Mukhopadhyay (1974) proposed a procedure for this problem when $F_\theta(x)$ is the exponential distribution function; his procedure is defined in section 2 together with a modified procedure which guarantees a coverage probability of at least $1 - \alpha$ for all $d > 0$. In section 3 two nonparametric procedures are

discussed.

The questions that arise are: How well do the asymptotic results describe the behaviour of the procedures? How do these procedures compare with one another? What happens with the procedures when $F_\theta(x)$ is not the exponential distribution? In section 4 these questions are answered by investigating the performances of the procedures in a Monte Carlo study.

2. PROCEDURES FOR THE TRUNCATION PARAMETER OF THE EXPONENTIAL DISTRIBUTION

Let $F_\theta(x)$ be the exponential c.d.f. with density function

$$g_{\theta,\sigma}(x) = \sigma^{-1} e^{-(x-\theta)/\sigma} I(x \geq \theta)$$

where both θ and σ are unknown and $I(\cdot)$ denotes the indicator function. Let $X_{n,1} \leq X_{n,2} \leq \dots \leq X_{n,n}$ denote the order statistics of $X_1, X_2, \dots, X_n$ and define

$$\hat{\sigma}_n = \sum_{i=2}^n (n-i+1)(X_{n,i} - X_{n,i-1})/n.$$

2.1 PROCEDURE OF MUKHOPADHYAY (1974)

Mukhopadhyay (1974) proposed the following interval:

$$I_{N_m} = [X_{N_m,1} - d, X_{N_m,1}]$$

where the sample size N_m is determined by

$$(2.1) \quad N_m = \inf\{n \geq n_0 \geq 2 \text{ for which } n \geq \hat{\sigma}_n(\log \alpha^{-1})/d\}.$$

It can be shown that

$$(2.2) \quad P(\theta \in I_{N_m}) \rightarrow 1 - \alpha \quad \text{as } d \rightarrow 0 \quad (\text{asymptotic consistency})$$

and

$$(2.3) \quad E(N_m) \sim \sigma(\log \alpha^{-1})/d \quad \text{as } d \rightarrow 0 \quad (\text{asymptotic efficiency}).$$

The drawback of this procedure is that it is only asymptotically consistent. For the construction of a fixed-width confidence interval for a normal mean when the variance is unknown, Simons (1968) proposed a procedure which is consistent for all $d > 0$.

2.2 A PROCEDURE WITH COVERAGE PROBABILITY AT LEAST $1 - \alpha$

Swanepoel and van Wyk (1982) proved that there exists a finite integer k such that if sampling is stopped according to (2.1) and k additional observations, $X_{N_m+1}, \dots, X_{N_m+k}$, are taken, then

$$(2.4) \quad P(\theta \in I_{N_m+k}) \geq 1 - \alpha \quad \text{for all } \theta, \sigma \text{ and } d.$$

Also

$$(2.5) \quad E(N_m+k) \sim \sigma(\log \alpha^{-1})/d \quad \text{as } d \rightarrow 0.$$

A question that arises is: what is the value of k ? In order to answer this question we follow the methods of Osborn and Simons (1970): Let the sequence $\{Y_n, n \geq 1\}$ be i.i.d. with density function

$$g_{0,1}(y) = e^{-y} I(y \geq 0).$$

Define $\bar{Y}_n = \frac{1}{n} \sum_{i=1}^{n-1} Y_i$ and a stopping variable

$$(2.6) \quad N_Y = \inf\{n \geq n_0 \geq 2 \text{ for which } n \geq \bar{Y}_n(\log \alpha^{-1})/d\}.$$

As in Swanepoel and van Wyk (1982) it can be shown that

$$(2.7) \quad P(\theta \in I_{N_m+k}) = E \left(1 - e^{-(N_Y+k)d/\sigma} \right) \\ = \sum_{n=n_0}^{\infty} (1 - e^{-(n+k)d/\sigma}) P(N_Y = n).$$

Now let

$$(2.8) \quad p_{n_0}(r, x) = \sum_{n=n_0}^{m_0} (1 - e^{-(n+x)r}) P(N_Y = n)$$

then $p_{n_0}(d/\sigma, k)$ approximates the probability in (2.7) for m_0 large.

In order to evaluate (2.8) the probability distribution of N_Y must be obtained. Using techniques similar to those of Robbins (1959) we have the following theorem:

Theorem: For $n_0 \geq 2$ and $a_n = n^2 d / \log \alpha^{-1}$ we obtain

$$P(N_Y > m) = \begin{cases} 1 & \text{for } m < n_0 \\ e^{-a_m} \left(\sum_{j=1}^{m-n_0+1} h_{m-j}(a_m) + \sum_{j=1}^{n_0-2} h_{n_0-1}^{(j)}(a_m) \right) & \text{for } m \geq n_0 \end{cases}$$

where

$$h_m(x) = \begin{cases} x^{n_0-2} / (n_0-2)! & \text{for } m = n_0-1 \text{ and } x > 0 \\ \sum_{j=1}^{m-n_0+1} h_{m-j}(a_m) (x-a_m)^j / j! \\ + \sum_{j=1}^{n_0-2} h_{n_0-1}^{(j)}(a_m) (x-a_m)^{m-n_0+j+1} / (m-n_0+j+1)! & \text{for } m = n_0, n_0+1, \dots \text{ and } x \geq a_m. \end{cases}$$

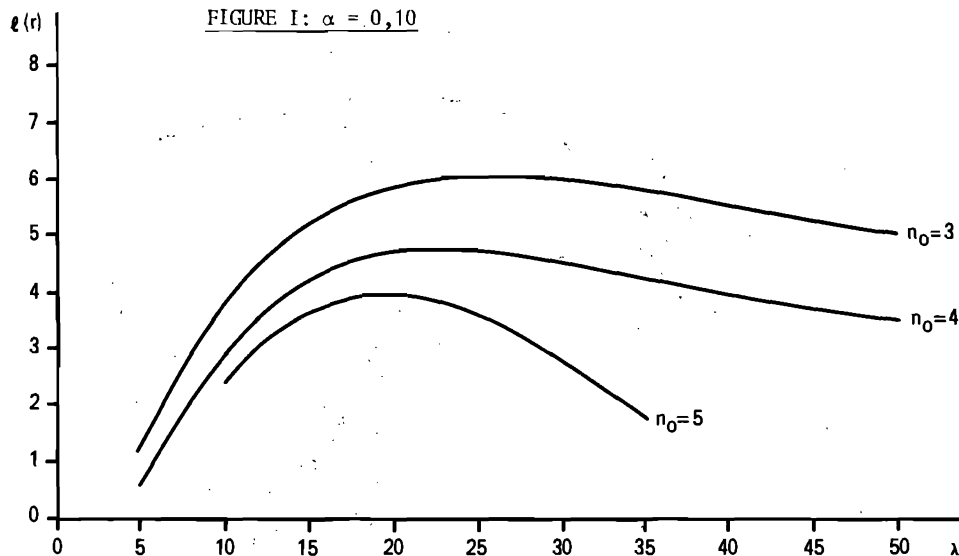
$(h_{n_0-1}^{(j)}(a_m))$ is the j -th order derivative of $h_{n_0-1}(x)$ evaluated in $x = a_m$.

Using this theorem we now proceed to obtain the magnitude of k . Let n_0 and α be fixed, then for a large number of values r a real number $\ell(r)$ can be calculated such that

$$p_{n_0}(r, \ell(r)) = 1 - \alpha.$$

In figures I, II and III these $\ell(r)$ -values are plotted against $\lambda(r) = (\log \alpha^{-1})/r$. From these figures the value k can be determined as the smallest integer larger than the maximum value of the graph. In table I the values of k are given for different values of $n_0 \geq 3$ and α . For $n_0 = 2$ the maximum value occurs at some point outside our computable range.

FIGURE I: $\alpha = 0,10$



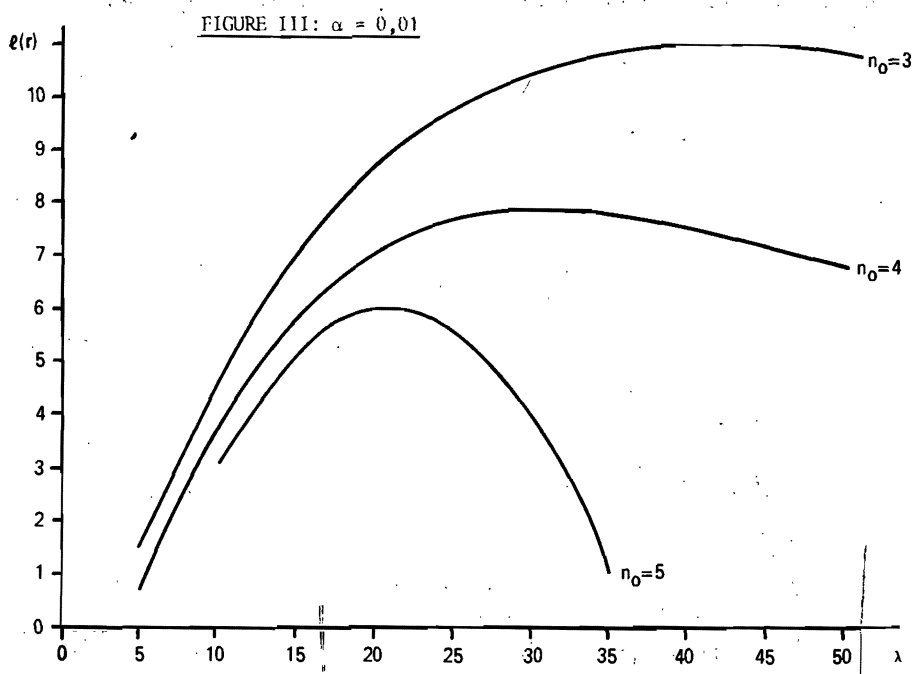
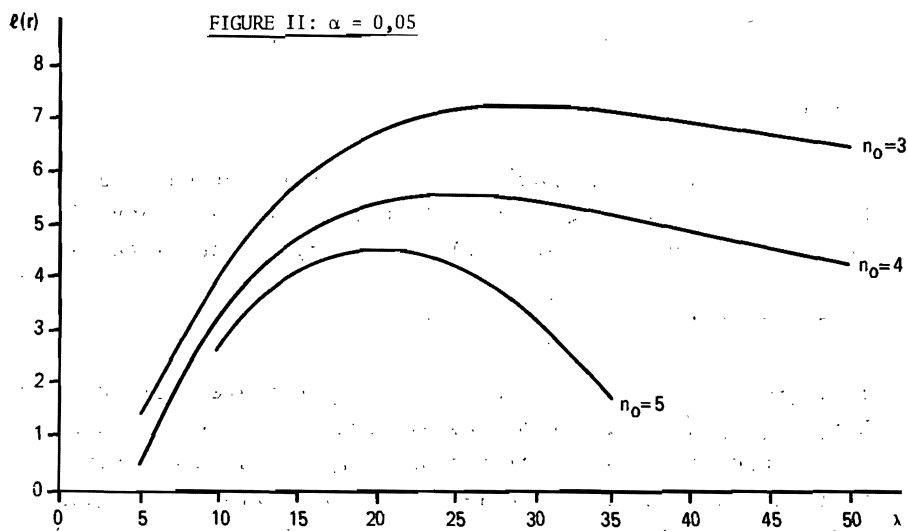


TABLE 1. VALUES OF k

n_0	α	0,1	0,05	0,01
3		7	8	11
4		5	6	8
5		4	5	6

3. NONPARAMETRIC PROCEDURES

In most practical situations the distribution function $F_\theta(x)$ is entirely unknown. For this case Swanepoel and van Wyk (1981(a)) proposed a sequential and a two-stage procedure:

Let $f_\theta(\theta) \equiv \lim_{x \rightarrow \theta} F'_\theta(x)$ and assume $f_\theta(\theta) > 0$, then under certain regularity conditions on $F_\theta(x)$ it can be shown (by using theorem 2.1.5 of Galambos (1978)) that

$$(3.1) \quad n(X_{n,1} - \theta) f_\theta(\theta) \xrightarrow{D} Y \quad \text{as } n \rightarrow \infty$$

where Y is exponentially distributed with density function $g_{0,1}(y) = e^{-y} I(y \geq 0)$. ($\xrightarrow{D}$ denotes convergence in distribution.) If $f_\theta(\theta)$ were known (3.1) can be used to find a sample size that leads to an interval with coverage probability approximately $1 - \alpha$ for n large. For $f_\theta(\theta)$ unknown (as is usually the case) the sample size cannot be determined and the following procedures are proposed: Let $\hat{f}_n$ be a strongly consistent non-parametric estimator of $f_\theta(\theta)$ (e.g. Swanepoel and van Wyk (1981(b))). Define the stopping variable

$$(3.2) \quad N = \inf\{n \geq n_0 \text{ for which } n \geq (\log \alpha^{-1}) / d\hat{f}_n\} \quad \text{and then}$$

take $I_N = [X_{N,1} - d, X_{N,1}]$ as interval for θ . It is shown by Swanepoel and van Wyk (1981(a)) that

$$(3.3) \quad P(\theta \in I_N) \rightarrow 1 - \alpha \quad \text{as } d \rightarrow 0$$

and

$$(3.4) \quad E(N) \sim (\log \alpha^{-1}) df_{\theta}(\theta) \quad \text{as } d \rightarrow 0.$$

In many practical situations a two-stage procedure is more convenient than a sequential procedure. Consider the following procedure:

$$n_0 = \max\{2, [(\log \alpha^{-1})/d]^{\frac{1}{2}} + 1\}.$$

($[x]$ denotes the integer part of x .) Compute $\hat{f}_{n_0}$ and define

$$(3.5) \quad N_0 = \max\{n_0, [(\log \alpha^{-1})/\hat{d}f_{n_0}] + 1\}.$$

The interval for θ is then $I_{N_0} = [X_{N_0,1} - d, X_{N_0,1}]$.

This procedure is motivated from Mukhopadhyay (1980) (see also Mukhopadhyay (1982)). From theorem 2 of Swanepoel and van Wyk (1981(a)) it follows that:

$$(3.6) \quad P(\theta \in I_{N_0}) \rightarrow 1 - \alpha \quad \text{as } d \rightarrow 0$$

and

$$(3.7) \quad E(N_0) \sim (\log \alpha^{-1})/df_{\theta}(\theta) \quad \text{as } d \rightarrow 0.$$

Remark: For the exponential distribution, $f_{\theta}(\theta) = 1/\sigma$ and it follows that

$$E(N)/E(N_m) \rightarrow 1 \quad \text{as } d \rightarrow 0$$

and

$$E(N_0)/E(N_m) \rightarrow 1 \quad \text{as } d \rightarrow 0.$$

5. MONTE CARLO STUDY

The Monte Carlo study was performed on an IBM 4331 computer of the Potchefstroom University. Uniform random variables were obtained by using the random number generator of the International Mathematical and Statistical Libraries (IMSL, 1980) and were then transformed.

In the Monte Carlo study the following distributions are used:

Exponential c.d.f. (EXP) with density function

$$g_{0,1}(y) = e^{-y}I(y \geq 0).$$

Normal c.d.f. truncated at 0 (TRN) with density

$$g(y) = (2/\pi)^{1/2}e^{-y^2/2}I(y \geq 0).$$

Uniform c.d.f. (U) with density function

$$g(y) = I(0 \leq y \leq 1).$$

The procedures are investigated for $n_0 = 5$, $\alpha = 0,05$ and $d = 0,300; 0,100; 0,060; 0,020$ and $0,006$. Each Monte Carlo result is based on 1000 repetitions and the $\pm$ values in the tables are one standard error. The coverage probability and expected sample size are estimated for each procedure. In this section we will use the following notation for the procedures:

M for the procedure of Mukhopadhyay

S for the procedure of section 2.2

NS for the nonparametric sequential procedure

NT for the nonparametric two-stage procedure.

In order to investigate the robustness properties of the procedures (see tables II and III) we used the k -values of table I for the S-procedure.

From table II below we see that in the exponential case the estimates are near $1 - \alpha$ for small values of d . For larger d values the M-procedure does not attain the required coverage probability while the other procedures do. In the uniform and truncated normal cases we see that for the M- and S-procedures the estimates are far away from $1 - \alpha$ while the nonparametric procedures still attain the required coverage probability. This shows that the M- and S-procedures are not suitable when one is not absolutely sure that the data is exponentially distributed.

For the M- and S-procedures we see from table III below that the estimates of the expected sample size are near the asymptotic values. For the NS-procedure the estimates are slightly larger than the asymptotic values and for the NT-procedure they are considerably larger. The reason for this is probably the fact that only the first n_0 observations are used to estimate $f_\theta(\theta)$; for the NS-procedure $f_\theta(\theta)$ is estimated *after each* observation, using all the observations available at each stage.

From tables II and III we see that the nonparametric procedures (NS and NT) compare very favourably with the S-procedure in the exponential case. The S-procedure is w.r.t. sample size slightly more efficient than the nonparametric procedures. The reason for this phenomenon is that the S-procedure uses the characteristics of the exponential distribution. However, for other distributions this procedure does not attain the required coverage probability and hence it does not make much sense to compare the sample sizes with those of the nonparametric procedures.

Concluding remarks: For the interval estimation of a truncation parameter we recommend using one of the nonparametric procedures since they have the required coverage probability and the expected sample sizes compare favourably with the M- and S-procedures. If the data is exponentially distributed and one has no doubt about this, then we recommend using the S-procedure.

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TABLE II. COVERAGE PROBABILITY, $\alpha = 0,05$
MONTE CARLO ESTIMATES

d	EXP	U	TRN
M-procedure			
0,300	0,887 $\pm$ 0,010	0,829 $\pm$ 0,012	0,792 $\pm$ 0,013
0,100	0,923 $\pm$ 0,008	0,664 $\pm$ 0,015	0,816 $\pm$ 0,012
0,060	0,934 $\pm$ 0,008	0,699 $\pm$ 0,015	0,839 $\pm$ 0,012
0,020	0,947 $\pm$ 0,007	0,759 $\pm$ 0,014	0,841 $\pm$ 0,012
0,006	0,947 $\pm$ 0,007	0,755 $\pm$ 0,014	0,845 $\pm$ 0,011
S-procedure			
0,300	0,978 $\pm$ 0,005	0,964 $\pm$ 0,006	0,943 $\pm$ 0,007
0,100	0,953 $\pm$ 0,007	0,800 $\pm$ 0,013	0,873 $\pm$ 0,011
0,060	0,952 $\pm$ 0,007	0,817 $\pm$ 0,012	0,859 $\pm$ 0,011
0,020	0,959 $\pm$ 0,006	0,784 $\pm$ 0,013	0,856 $\pm$ 0,011
0,006	0,951 $\pm$ 0,007	0,802 $\pm$ 0,013	0,861 $\pm$ 0,011
NS-procedure			
0,300	0,973 $\pm$ 0,005	0,970 $\pm$ 0,005	0,947 $\pm$ 0,007
0,100	0,962 $\pm$ 0,006	0,940 $\pm$ 0,008	0,941 $\pm$ 0,007
0,060	0,965 $\pm$ 0,006	0,935 $\pm$ 0,008	0,940 $\pm$ 0,008
0,020	0,948 $\pm$ 0,007	0,948 $\pm$ 0,007	0,956 $\pm$ 0,006
0,006	0,965 $\pm$ 0,006	0,948 $\pm$ 0,007	0,952 $\pm$ 0,007
NT-procedure			
0,300	0,989 $\pm$ 0,003	0,992 $\pm$ 0,003	0,985 $\pm$ 0,004
0,100	0,986 $\pm$ 0,004	0,969 $\pm$ 0,007	0,970 $\pm$ 0,005
0,060	0,975 $\pm$ 0,005	0,972 $\pm$ 0,005	0,956 $\pm$ 0,006
0,020	0,967 $\pm$ 0,006	0,954 $\pm$ 0,007	0,961 $\pm$ 0,006
0,006	0,969 $\pm$ 0,005	0,959 $\pm$ 0,006	0,952 $\pm$ 0,007

TABLE III. EXPECTED SAMPLE SIZE

d	EXP		U		TRN	
	Asymp. value	M.C. value	Asymp. value	M.C. value	Asymp. value	M.C. value
M-procedure						
0,300	10,0	8,5±0,1	5,0	5,1±0,01	8,0	6,6±0,1
0,100	30,0	27,8±0,2	15,0	12,2±0,1	23,9	21,6±0,2
0,060	50,0	48,3±0,2	25,0	22,6±0,1	39,9	38,2±0,2
0,020	150,0	149,8±0,4	75,0	73,1±0,2	119,7	118,3±0,2
0,006	500,0	500,4±0,7	250,0	248,4±0,3	398,9	397,1±0,5
S-procedure						
0,300	10,0	13,4±0,1	5,0	10,1±0,01	8,0	11,7±0,1
0,100	30,0	33,2±0,2	15,0	17,0±0,1	23,9	26,7±0,2
0,060	50,0	53,5±0,2	25,0	27,8±0,1	39,9	43,1±0,2
0,020	150,0	154,1±0,4	75,0	78,2±0,2	119,7	123,4±0,3
0,006	500,0	503,5±0,7	250,0	253,7±0,3	398,9	402,0±0,5
NS-procedure						
0,300	10,0	14,8±0,1	10,0	12,5±0,1	12,5	14,6±0,1
0,100	30,0	37,0±0,3	30,0	32,8±0,3	37,5	40,2±0,3
0,060	50,0	59,9±0,5	50,0	52,6±0,4	62,5	65,6±0,5
0,020	150,0	168,0±0,9	150,0	152,9±0,8	187,7	190,8±1,0
0,006	500,0	537,1±2,0	500,0	504,3±1,9	625,0	631,7±2,4
NT-procedure						
0,300	10,0	27,1±0,5	10,0	20,1±0,2	12,5	24,5±0,4
0,100	30,0	62,7±1,1	30,0	44,8±0,5	37,5	59,3±0,9
0,060	50,0	93,7±1,5	50,0	77,5±1,0	62,5	91,3±1,3
0,020	150,0	238,3±3,2	150,0	187,0±1,8	187,7	241,2±2,8
0,006	500,0	699,4±7,6	500,0	590,6±5,4	625,0	734,8±7,2

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